Birla Institute of Technology and Science
I semester 2011-2012
EEE C424/INSTR C313, Microelectronic Circuits
Comprehensive examination (closed-Book)
Total Time: 3 hrs. , M.M = 100

Instructions-

- State your assumptions clearly. All symbols have their usual meaning. **Question paper has total - 6- questions**
- **All parts of the same question should be done together.**. Marks will be deducted for wrong units. Your answers must be justified with proper reasons.
- Sketches should be properly labeled. Marks would be deducted for calculation mistakes. Use subscripts to distinguish between different transistors
- Take $\gamma=\lambda=0$ in drain current calculations unless necessarily required or specified in the question.
- Wherever not shown, take bulk of NMOS connected to ground and bulk of PMOS connected to Vdd.
- Use square law for MOS transistor. Ignore wiring capacitance. If not specified, assume all transistors are biased in active region,

Unless mentioned in questioned specifically, assume all MOS transistors are biased in saturation
Unless Given use -- $V_{DD} = 3.3V$,
for NMOS device:
$\mu_n Cox = 140\ \mu A/V^2$, $V_T = 0.7$ V , $\lambda = 0.2\ V^{-1}$, Cox= 0.38pF/um^2 , Lmin= 1um
for PMOS device :
$\mu_p Cox = 40\ \mu A/V^2$, $V_T = -0.8$ V , $\lambda = 0.2\ V^{-1}$, Cox =0.38 pF/um^2 , Lmin= 1um

Q1. Consider the circuit of fig. 1. Assume all transistors have overdrive voltage= 0.2V

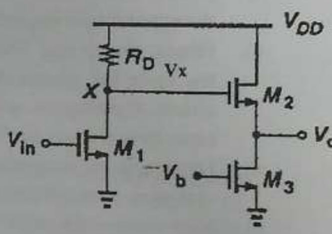


Fig 1

a) Derive the expression for total low frequency gain (vout/vin) without any approximation
b) Derive the expression of capacitive load at node x and out.
c) Estimate the value of d.c voltage Vout to set Vx= ½Vdd .
d) Recalculate the d.c voltage Vx for Vout = ½Vdd
e) Now, sketch and label the maximum possible voltage Vx (t) vs. t , and Vout (t) vs. t for part (c) and part (d) for input to be a symmetrical sinusoid with amplitude 100mV.
f) Now, determine limit on maximum possible dc gain of the entire circuit using results of part (c), part (d), and part (e), part (f)

2+4+1+1+8+2=18

Q2. For the circuit of fig. 2, assume all transistors have Voverdrive = 0.2V. Now due to PVT variations, small signal parameters change by ±10% Assuming γ=0

a) Determine the approximate value of d.c gain (vout2/vin)

b) Determine the value of w_{-3dB} frequency intuitively for total capacitive load of value 1pF and 0.1 pF at nodes out1, and out2 respectively

c) Determine the variation in unity gain bandwidth due to PVT variations

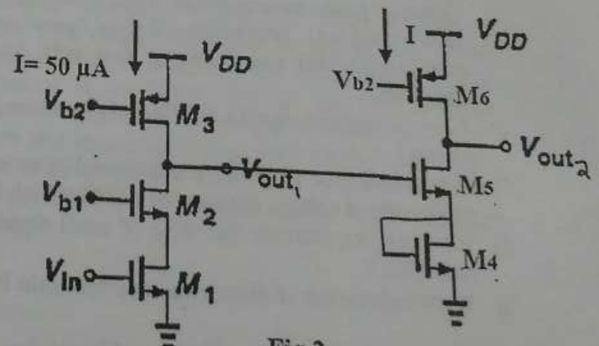


Fig 2

4+4+8=16

Q3. Consider the circuit shown in fig. 3. Vin= vin1-vin2, Vov for all transistors= 0.2V

a) Sketch, label and design the circuit for the box shown in given circuit. Hence determine the minimum voltage requirement at node P

b) Determine the low frequency small signal gain (Vout/ Vin), and (V_F/ vin).

c) Determine input common mode range, output common mode range, output voltage signal swing at Vout node, voltage signal swing at node F

d) Show the location of Q point of the input transistors on (I_D vs. V_{DS}), and (I_D vs. V_{GS}) plot.

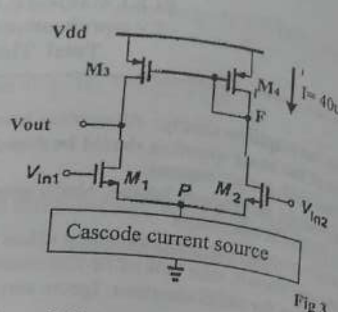


Fig 3

5+4+4+6=19

Q4. For the fig 4, given Iref=I=100µA, $(Cgd)_{M6}$ =10fF, neglect Cgd of other transistors. Also neglect C_{db}, C_{sb}, of all Vov for all transistors= 0.2V, Q8, Q5, Q7 are matched.

a) Construct a labelled high frequency trans-conductance amplifier model of the circuit.

Without Cc in the circuit--

b) Determine the number of pole frequencies and derive the approximate expression and their values intuitively

c) Hence, sketch and label Bode Magnitude plot (not to the scale but should be qualitatively correct) for the circuit showing dc gain, location of poles and gain cross over frequency. Neglect transmission zeros.

d) Determine an inequality to express phase margin and comment on stability of OP-AMP.

e) Now discuss the need of Cc to achieve stability of the circuit.

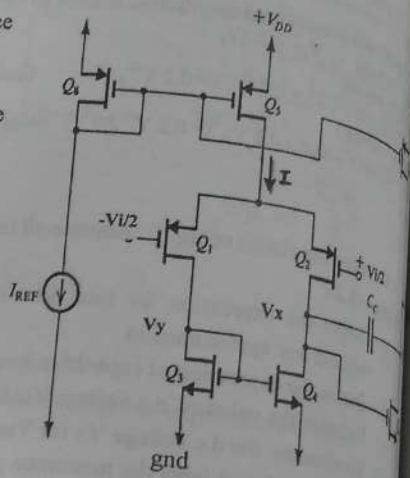


Fig 4

4+10+5+1+...

Q5. Answer the following in brief (*you may draw figures to bring clarity in your answer*)

a) Give the technical name of a current sampling current mixing feedback amplifier

b) A fully differential amplifier with complementary input applied can operate with diode connected transistor implementing Iss. Justify

c) Give two prime advantages of using feedback in design of amplifiers

d) How would you compensate a 2 stage operational amplifier (differential pair followed by common source amplifier) which has dominant first pole frequency contributed by output node of common source amplifier?

e) How is negative feedback effect different in Wilson current mirror in comparison to cascode current mirror?

f) A common gate amplifier is configured in series shunt feedback amplifier. Can it become unstable when phase output voltage changes by ±180° at high frequencies? Justify

g) How can you increase the value of small signal parameter 'ro' without changing the bias current through MOSFET?

h) Write expression of Short circuit current gain frequency of the n channel MOSFET?

8x2...

Q6. Design a common source amplifier with resistive load with a dc gain of 10, dc power consumption=132... Vov= 0.2v Now use a cascade of two above designed common source amplifier as feed forward amplifier... configure a series shunt feedback amplifier with feedback factor β= 1/100 mho without changing dc gain of the forward amplifier. Sketch and label the complete schematic of your feedback amplifier

Birla Institute of Technology and Science, Pilani
EEE F244 / INSTR F244 Microelectronic Circuits
Comprehensive examination II Semester 2014-2015
CLOSED BOOK Max. Marks=110

Time: 3 HRS Date: 6-5-2015

Unless Given specifically
Take -- V_{DD} = 3.3 V, Lmin=1um
For NMOS device $\mu_n Cox = 140\ \mu A/V^2$, $V_T = 0.7$ V, $\lambda = 0.1\ V^{-1}$, $Cox = 38 pF/um^2$
For PMOS device $\mu_p Cox = 40\ \mu A/V^2$, $V_T = -0.8$ V, $\lambda = 0.1\ V^{-1}$, $C_{ox} = 38\ pF/um^2$

NOTE

If not specified in question,——

- ignore γ, λ in drain current equation
- Bulk of nmos connected to ground and bulk of pmos connected to Vdd. Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET are biased in saturation region. Label your sketches properly. All symbols have usual meaning. Assume matching of transistors wherever necessary.

Q.1(a) Consider a MOSFET carrying a drain current (I_D) of 1mA with $V_{DS} = 0.5$ V in active region. Determine the change in I_D if V_{DS} rises to 1V and $\lambda = 0.1\ V^{-1}$. What is device output impedance 'ro'? Assuming $\lambda \propto 1/L$, calculate the change in I_D and output impedance if both Width and Length are doubled.

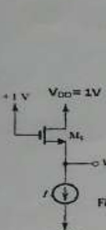


Fig.(1)

Q1(b) For the **Fig.(1)**, $V_{GS} = V_{DS} = 3V$. use $\lambda = 0.1\ V^{-1}$

(i) Find the voltage V_2, if the transistor is in saturation at $I = I_D = 2mA$ and $V_T = 1V$.
(ii) Determine minimum value of drain -source voltage of M1 to remain in saturation region.
(iii) Now a resistor R is inserted between Vdd and drain terminal of M1. For result of part (ii), find the value of R
(iv) using above results, If current source 'I' requires a voltage of 2V across it, determine the value of r that can be inserted between source terminal of M_1 and current source. [20]

Q-2 Consider the circuits given in **Fig(2)**.

a) Redraw the given circuit in **Fig. 2(a)** for very low frequency operation. Find the expression of voltage gain (V_{out}/V_{in}) for the new circuit intuitively.
b) Redraw the circuit in **Fig. 2(a)** for very high frequency operation. Draw small signal model of new circuit and find expression of voltage gain (V_{out}/V_{in}). ignore parasitic capacitances.
c) The circuit shown in **Fig. 2(b)** has a bias current (I_{bias}) of 100μA and W/L values for all the transistors are 10. Calculate values of small signal voltage gain (V_{out}/V_{in}) and output voltage swing. Assume $\gamma = 0$ and $\lambda \neq 0$. [20]

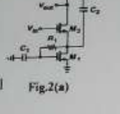


Fig.2(a)

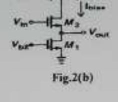


Fig.2(b)

Q-3 For the circuits shown in **Fig.3 (a, b, c, d, e, f g, h, i, j)**, assume all transistors have same g_m and r_o, I_{bias}, W/L, and uC_{ox}, V_{ov}. Ignore γ

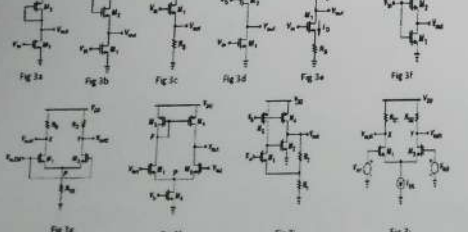


Fig 3a Fig 3b Fig 3c Fig 3d Fig 3e Fig 3f
Fig 3g Fig 3h Fig 3i Fig 3j

Answer the following to the point (very brief) *(please answer in same sequence)*-

a) For **fig. 3a**, write expression for d c voltage at node vout
b) For **fig. 3b**, write the expression for resistance at node vout.
c) For **fig. 3c**, write the expression for dc voltage (Vin-Vout)
d) For **fig 3d**.,write the expression for OCMR (OUTPUT COMMON MODE RANGE)
e) For **fig. 3g**, write the expression for ICMR (INPUT COMMON MODE RANGE)
f) For **fig. 3g**, sketch and label common mode half circuit.
g) For **fig. 3h**, write the expression for output voltage swing
h) For **fig. 3f**, write the expression for Gm
i) For **fig. 3e**, write the expression for resistance at node Vout
j) For **fig. 3d**, write the expression for low frequency small signal voltage gain.
k) Re-evaluate part (j) if voltages Vin and Vb are swapped
l) identify type of feedback in **fig 3i**
m) For **fig. 3j**, write the expression for voltage swing in differential mode of operation
n) Write the expression for feedback factor β in **fig 3i**.
o) For **fig 3h**, write the expression for dc voltage at node Vout
p) For **fig. 3e**, write the expression for Gm.

[30]

Q4.(a):): In the system shown in figure 4 a , the System on Chip and peripheral circuitry operates at different voltage levels. In order to make signals level compatible with each other interface circuits are required as shown in given figure.

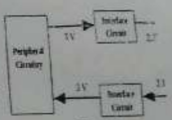
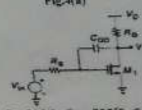


Fig.4(a)

Assume I_{bias}= 0.5mA is the current flowing through each interface circuit also values of $\mu_n C_{ox}$ =134 $\mu A/V^2$ $\mu_p C_{ox}$=38 $\mu A/V^2$, Vt_n= 0.7V, Vt_p= - 0.7V, V_{dd}=3V. Wherever required, assume basic current mirror load with V_{ov}=0.3V.

a) Choose the appropriate amplifier circuits as per the requirement shown in figure. Write its name.
b) Replace the both interface circuits blocks with the transistor level implementations of amplifier.
c) Now design your chosen amplifier as per the specification required.



Q4.(b): For the circuit given in figure 4(b), consider following parameters : λ=0, Rs=200Ω, Cgs=250fF, Cgd
Cdb=100fF, g_m=150 Ω^{-1}, R_D=2KΩ.

a) Calculate, low frequency voltage gain (Vout/Vin) of the amplifier
b) Also, calculate the number of poles and zeros frequency intuitively, and identify the dominant pole
c) Now, plot the frequency response (both magnitude and phase plots) up-to 10^{15} rad/s of the amplifier with appropriate labeling.(not to the scale but qualitatively correct)

[20]

Q5: The amplifier circuit shown in **figure 5**, has the current source I_{SS} of 200μA with very high R_{SS} resistance. It requires the minimum voltage of 0.2 V. Given: $(W/L)_{1-10}$ = 100, calculate the values of following using the intuitively (Assume γ=0) , I (M_9)= 100uA

(i) Input common mode range (ICMR)
(ii) Overall output voltage swing at V_{out1}.
(iii) Overall low frequency differential voltage gain (Ad=V_{out2}/V_{in}) using intuitive method.
(iv) Assuming the unity feedback, determine frequency (in rad/sec) for which phase margin is 45^0.
(Given: The sum of all parasitic capacitances at node 'V_{out1}' is C_{Vout1}=200 fF, sum of all parasitic capacitances at node x/ x' or y/y' node is 10 fF, the external capacitance C_L=500 fF). Also draw approximate Bode magnitude plot assuming a 2 dominant pole system (not to the scale but qualitatively correct
(v) If a compensating capacitor of C_c=0.5 nF , is connected between nodes 'V_{out1}' & node 'V_{out2}' then find the new locations of dominant poles of part (iv).
(vi) At the frequency obtained in part (iv), determine the new pahse angle

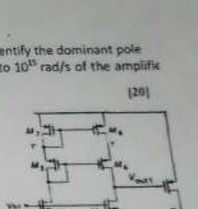


Fig.5

[20]

ompree Sol.

Q. 1 (a)

$I_{D1} = 0.5\,\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{th})^2(1+\lambda V_{DS1})$

$I_{D2} = 0.5\,\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{th})^2(1+\lambda V_{DS2})$

$I_{D2} = I_{D1} \frac{(1+\lambda V_{DS1})}{(1+\lambda V_{DS2})} = 1.048\,mA$

Change in $I_D = 48\,\mu A$. [4M]

$r_o = \frac{\Delta V_{DS}}{\Delta I_D} = 10.4\,k\Omega$ [1M]

(b) λ drops to $0.05\,V^{-1}$

$I_{D2} = 1.024\,mA$

$\therefore \Delta I_D = 24\,\mu A$ [4M]

$r_o = 20.84\,k\Omega$ [1M]

19 Q.1(b)

(i) $V_G = V_S = V_{DD} - V_{GS} = 1-3 = -2V$ [2M]

(ii) $V_{DS} = V_{GS} - V_T$, at the edge of saturation

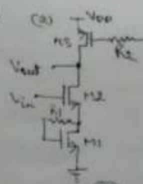
$V_{DS_{max}} = 3-1 = 2V$ [2M]

(iii) $R = \frac{1}{0.2\,mA} = 0.5\,k\Omega$ [as $V_S = V_G = -2V$] [3M]

(iv) $V_D = -9+2 = -7V$, $V_S = V_G - V_{GS} = 1-3 = -2V$

$R = \frac{-2-(-7)}{2\,mA} = 2.5\,k\Omega$ [3M]

Q2. (a)



(1M)

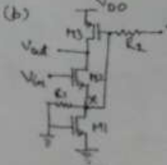
$\frac{V_{out}}{V_{in}} = G_m R_{out}$

$= \frac{g_{m2}}{1+g_{m2}\frac{1}{g_{m1}}} \cdot (r_{o3} \| \ldots)$

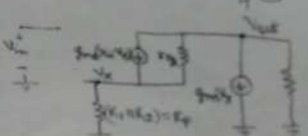
$\frac{g_{m2}}{1+\frac{g_{m2}}{g_{m1}}} \cdot (r_{o3} \| \ldots)$

$\approx \frac{g_{m2}}{1+\frac{g_{m2}}{g_{m1}}} \cdot (r_{o3} \| r_{o2}\ldots)$

(b)



(1M)



Applying KCL at node V_x & V_{out}

$\frac{V_x}{R_P} = g_{m2}(V_{in} - V_x) + \frac{V_{out} - V_x}{r_{o2}} = -g_{m3}\ldots$

$\frac{V_x}{R_P} = -\left(g_{m3}V_x + \frac{V_{out}}{r_{o3}}\right) \Rightarrow \frac{V_{out}}{V_x} = -r_{o3}\left(\frac{1}{R_P} + g_{m3}\right)$

$\frac{V_x}{R_P} = g_{m2}V_{in} - g_{m2}V_x + \frac{V_{out}}{r_{o2}} - \frac{V_x}{r_{o2}}$

$V_x\left(\frac{1}{R_P} + g_{m2} + \frac{1}{r_{o2}}\right) = g_{m2}V_{in} + \frac{V_{out}}{r_{o2}}$ — (2)

Using (1) & (2)

$-\frac{V_{out}}{r_{o3}\left(\frac{1}{R_P}+g_{m3}\right)}\left(\frac{1}{R_P}+g_{m2}+\frac{1}{r_{o2}}\right) = g_{m2}V_{in} + \frac{V_{out}}{r_{o2}}$

$-V_{out}\left(\frac{1}{r_{o2}} + \frac{\frac{1}{R_P}+g_{m2}+\frac{1}{r_{o2}}}{r_{o3}\left(\frac{1}{R_P}+g_{m3}\right)}\right) = g_{m2}V_{in}$

$\frac{V_{out}}{V_{in}} = \frac{-g_{m2}}{\frac{1}{r_{o2}} + \frac{\frac{1}{R_P}+g_{m2}+\frac{1}{r_{o2}}}{r_{o3}\left(\frac{1}{R_P}+g_{m3}\right)}} = \frac{-g_{m2} \cdot r_{o2} \cdot r_{o3}\left(\frac{1}{R_P}+\ldots\right.}{r_{o3}\left(\frac{1}{R_P}+g_{m3}\right) + r_{o2}\left(\frac{1}{R_P}\ldots\right.}$

2(c) $(W/L)_{1,2,3} = 10$

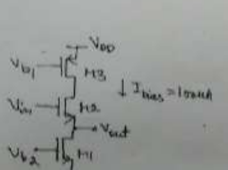


Fig 2b

R_{up} : resistance looking at source of M2

$$R_{up} = \frac{r_{o2} + r_{o3}}{1 + g_{m2} r_{o2}}$$

$R_{down} = r_o$ (R_{down} : resistance looking at drain of M1)

$$\frac{V_{out}}{V_{in}} = \frac{r_{o1}}{r_{o1} + \frac{r_{o2} + r_{o3}}{1 + g_{m2} r_{o2}}}$$

$$r_{o1} = r_{o2} = r_{o3} = \frac{1}{\lambda I_D} = \frac{1}{0.\times 100u} = 100k\Omega = 10^5 \Omega$$

($\because \lambda$ is same)

$$g_{m2} = \sqrt{2 I_D \mu_n C_{ox} \frac{W}{L}} = \sqrt{2 \times 100u \times 140u \times 10} = 0.529 \text{ mA/V}$$

20

$$\frac{V_{out}}{V_{in}} = \frac{10^5}{10^5 + \frac{2\times 10^5}{1 + 0.529m \times 10^5}} = 0.964 \text{ V/V}$$

- (4M)

Upper swing $= V_{DD} - |V_{ov2}| - |V_{ov3}|$

Lower swing $= V_{ov1}$

$$V_{ov2} = \frac{2 I_D}{g_{m2}} = \frac{2 \times 100u}{0.529m} = 0.378 \text{ V}$$

$$g_{m3} = \sqrt{2 I_D \mu_p C_{ox} \frac{W}{L}} = \sqrt{2 \times 100u \times 40u \times 10} = 0.283 \text{ mA/V}$$

$$V_{ov3} = \frac{2 I_D}{g_{m3}} = \frac{2 \times 100u}{0.283m} = 0.707 \text{ V}$$

$$V_{ov1} = \frac{2 I_D}{g_{m1}} = 0.378 \text{ V}$$

Upper swing $= 3.3 - 0.378 - 707 = 2.215$ V - (2M)

Lower swing $=$ ~~[illegible]~~ 0.378 V

- (2M)

Q.3 Answer

(a) fig 3(a) d.c voltage at node $V_{out} = V_{DD} - V_{SG_{M2}}$ (2)

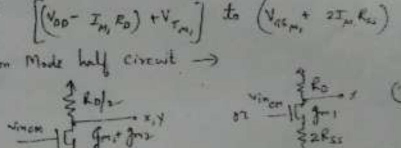
(b) fig 3b Resistance at node V_{out} $R_{out} = \frac{1}{g_{m2}} || r_{o2} || r_{o1} \approx \frac{1}{g_{m2}}$ (

(c) d.c voltage at node ($V_{in} = V_{out}$) fig 3(c) $= V_{GS_{M1}} = \sqrt{\frac{2 I_{M1}}{\mu_n C_{ox} \frac{W}{L}}}$

(d) fig 3d, OCMR $(V_{DD} - V_{ov_{M2}})$ to $V_{ov_{M1}}$ (2)

(e) fig 3g, ICMR $[(V_{DD} - I_{M1} R_D) + V_{T_{M1}}]$ to $(V_{GS_{M1}} + 2 I_{M1} R_{ss})$

(f) fig 3g Common Mode half circuit →



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(g) fig 3h output voltage swing $(V_{DD} - V_{ov_4})$ to $(V_{ov_{M2}} + \frac{I_{ss} R_s}{2})$

(h) fig 3f, $G_m =$ ~~$(g_{m1} + g_{m2})$~~ 0 zero

(i) fig 3e Resistance at node V_{out} → $R_{out} = R_D || [g_{m1} r_{o1} R_s + r_{o1}$

(j) fig 3d low freq. small signal voltage gain $= -g_{m1}$ (r

(k) fig(3d) V_{in}, V_b swapped, low freq. small signal voltage gain $=$ -

(l) type of feedback in fig 3i → voltage sampling, voltage (series - shunt) ff

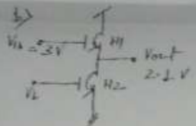
(m) 3j voltage swing (differential) $=$ ~~[illegible]~~ $\{V_{DD}$ to V_{ov_m}

(n) β (feedback factor) $= \frac{V_{out}}{V_f} = \frac{R_1}{R_1 + R_2}$, (fig 3i) (

(o) fig 3h, Expression for d.c voltage at node V_{out} =

g_{m1} (2)

b)



$$\left(\frac{W}{L}\right)_1 = \frac{2 \times 0.5 \times 10^{-3}}{134 \times 10^{-6}(0.9-0.7)^2}$$

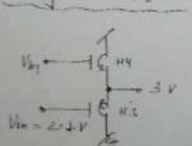
$$\left(\frac{W}{L}\right)_1 = 186$$

2

$$\left(\frac{W}{L}\right)_2 = \frac{2 \times 0.5 \times 10^{-3}}{134 \times 10^{-6}(0.3)^2} = 82.9$$

2

Interface circuit 2



$$\left(\frac{W}{L}\right)_3 = \frac{2 \times 0.5 \times 10^{-3}}{38 \times 10^{-6}(0.9-0.7)^2} = 657$$

2

$$\left(\frac{W}{L}\right)_4 = \frac{2 \times 0.5 \times 10^{-3}}{134 \times 10^{-6} \times (0.3)^2} = 82.9$$

2

Total = 10M

a) $A_v = \frac{V_{out}}{V_{in}} = g_m \times r_L$ | $r_{out} \approx r_L \| r_o \approx r_L$

$= 150 \times 2 \times 10^{3}$

$= 300000$

2M $A_v \cong 109\ dB$

b) 2 poles, 1 zero

$$\omega_{p1} = \frac{1}{R_S(C_{gs} \| A \times C_{gd})}$$

$\omega_{p1} = 0.554 \times 10^{6}$ rad/sec 1

$$\omega_{p2} = \frac{1}{R_L(C_{gd} + C_{db})}$$

$\omega_{p2} = 2.77 \times 10^{6}$ rad/sec 1

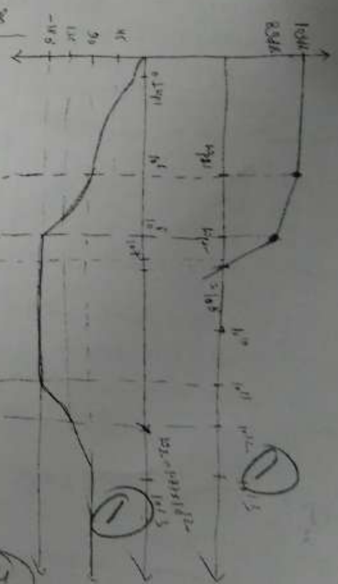
$$\omega_z = \frac{g_m}{C_{gd}} = 1.871 \times 10^{12} \text{ rad/sec}$$

1

$\omega_{p1} = 0.55 \times 10^{6}$ rad/sec

$\omega_{p2} = 2.77 \times 10^{6}$ rad/sec

$\omega_z = 1.87 \times 10^{12}$ rad/sec



Total ⇒ 10M

21

Q5:

$I_{(M_1)} = I_{(M_2)} = \frac{I_{SS}}{2} = 0.1\ mA$

$r_{on} = r_{op} = \frac{1}{\lambda I_D} = 100\ K\Omega$

$V_{ov,n} = \sqrt{\frac{2 I_{D(Mn)}}{\mu_n C_{ox} W/L}} = \sqrt{\frac{2 \times 0.1 \times 10^{-3}}{140 \times 10^{-6} \times 100}}$

$= 0.119\ V$

$V_{ov,p} = \sqrt{\frac{2 I_{D(Mp)}}{\mu_p C_{ox} W/L}} = \sqrt{\frac{2 \times 0.1 \times 10^{-3}}{40 \times 10^{-6} \times 100}}$

$= 0.223\ V$

$g_{mn} = \frac{2 I_D}{V_{ov,n}} = 1.68\ mA/V$

$g_{mp} = \frac{2 I_D}{V_{ov,p}} = 0.896\ mA/V$

22

(i) $V_{in(min)} = V_{ov(I)} + V_{GS(M1)}$

$= 0.2 + (0.7 + 0.119) = 1.019\ V$

$V_{in(max)} = V_{DD} - 2|V_{tp} + V_{op}| + V_{tn}$

$= (-2 \times 1.023) + 0.7$

$= 1.954\ V$

$ICMR = 0.935\ V$ (3M)

(ii) Upper swing $= V_{DD} - |V_{tp} + 2V_{ovp}|$

$= 3.3 - |0.8 + 2 \times 0.223| = 2.054\ V$

lower swing $= V_{ov(I)} + 2V_{ov,n}$

$= 0.2 + 2 \times 0.119 = 0.438\ V$

overall swing $= 1.616\ V$ (3M)

(iii) $A_d = A_1 \times A_2$

$= g_{mn} (g_{mn} r_{on}^2 \,||\, g_{mp} r_{op}^2)$

$A_d = 1.68 \times 10^{-3} [16.8M \,||\, 18.96$

$= 1.68 \times 10^{-3} \times 5.84M = 9823$

$A_2 = g_{mp} [r_{on} || r_{op}]$

$= 0.896 \times 10^{-3} \times 50K = 44.8$

$A_d = A_1 \times A_2 = 4.4 \times 10^5$ (4M)

(iv) $R_{out1} = 5.84\ M\Omega$, $C_{out1} = 2$

$R_{out2} = 50K$, $C_L = 500\ fF$

$\omega_{p1} \cong \frac{1}{R_{out1} \times C_{out1}} = 0.856\ Mrad$

$(0.1362\ MHz)$

$\omega_{p2} \cong \frac{1}{R_{out2} \times C_L} = 40\ M\ rad/sec$

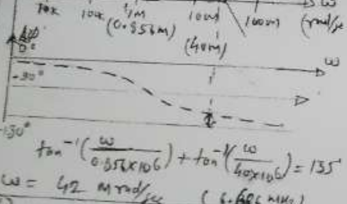
$(6.36\ MHz)$ (5M)

$\omega_{px} \cong \omega_{py} \cong \frac{1}{R_x C_x} = \frac{1}{R_y C_y}$

$\cong \frac{1}{100K \times 10fF} = 1000\ Mrad/sec$

$(159.15\ MHz)$

$A(j\omega) = \frac{4.4 \times 10^5}{\left(1 + \frac{j\omega}{0.856 \times 10^6}\right)\left(1 + \frac{j\omega}{40 \times 10^6}\right)}$



$\tan^{-1}\left(\frac{\omega}{0.856 \times 10^6}\right) + \tan^{-1}\left(\frac{\omega}{40 \times 10^6}\right) = 135$

$\omega = 42\ M\ rad/sec$ $(6.686\ MHz)$

(v) $\omega_{p1} = 7.41\ Krad/sec$ $\{ C_{out1} = 23.1\ fF$, $C_{out2} = 1\ pF$

$\omega_{p2} = \frac{1}{(r_{on} || 1/g_{mp}) C_{out2}} = 0.895\ Grad/sec$ 3M

(vi)